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Human Kinetics
ORIGINAL INVESTIGATION

Simultaneous Upper- and Lower-Limb Postactivation Performance Enhancement After Clean and Jerk

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Studies on postactivation performance enhancement (PAPE) have used different exercises as a conditioning activity to investigate potentiation, but exclusively in upper limbs (UL) or lower (LL) limbs, or contralateral potentiation. A single exercise capable of inducing PAPE in both UL and LL is currently unknown. The present study explored the effect of the clean and jerk (C&J) as a conditioning activity for simultaneously producing PAPE interlimbs at the fourth, seventh, and 12th minutes postintervention. Twelve male weightlifters with 1-repetition maximum (1RM) in the C&J equivalent to $\geq 1.15 \times$ body mass were randomly submitted to 2 experimental conditions (C&J and control [CON]). The C&J condition consisted of general warm-up (running on a treadmill and self-selected preparatory exercises) and 4 sets of 3 repetitions of C&J with 2 minutes between them (30%1RM, 50%1RM, 65%1RM, and 80%1RM) followed by a countermovement jump and a bench-press throw on a Smith machine after 4, 7, and 12 minutes, to measure the magnitude of PAPE in UL and LL. No previous exercise preceded countermovement-jump and bench-press-throw tests in the CON besides general warm-up. The main finding was that, regardless of time, the C&J resulted in greater height on countermovement jump and Smith machine bench-press throw when compared with the CON, presenting a similar effect size between UL and LL (34.6 [3.9] vs 33.4 [4.1] cm [+3.66%]; $P = .038$; effect size = 0.30 and 30.3 [4.7] vs 29.0 [5.1] cm [+4.44%]; $P = .039$; effect size = 0.26), respectively. Thus, C&J can be useful to produce PAPE simultaneously among members.

Keywords: warm-up, strength, countermovement jump, power, bench-press throw

Simultaneous enhancement in muscle strength and power in both limbs is critical to success in numerous sporting activities (eg, throwing and jumping).[1,2] Precisely, literature has reinforced the need for more studies that bring practical contexts (eg, training programs and protocols) with a focus on improving sports performance. In particular protocols that are not directed to just one specific sport but that can be used for different modalities. Postactivation performance enhancement (PAPE) is a phenomenon that can acutely enhance performance.[3] Apparently, neural factors such as greater recruitment of motor units and excitability of motoneurons, as well as greater muscle fiber conduction velocity, are the most preponderant to generate PAPE.[4,5] The momentary change in the pennation angle of the muscle fibers also seems to be decisive in this process.[6] In addition, some factors that may optimize PAPE effects are (1) individual strength level,[4] (2) adequate rest intervals between the conditioning activity (CA) and PAPE manifestations (ie, time between the end of the CA and the beginning of the main task),[7,8] and (3) CA: intensity[8] and exercise selection.[5,9] A meta-analysis indicated that for individuals with high levels of muscle strength, CA with intensity ranging from 60% to 84% one-repetition maximum (1RM) combined with a 7-minute rest interval before a main activity is recommended,[8] although some practitioners may also benefit from the PAPE effect with different load zones and intervals.[10] Furthermore, the biomechanical similarity between the exercise selected for CA and the main activity seems to be an important factor in the PAPE effect. Since, normally, the CA used in studies aims to generate local effect in the limb (upper limbs—UL or lower limbs—LL) and in the muscular chain involved, as well as specifically to the vectors of the movement that these muscles act on.[3,9–19] Thus, depending on the mechanical characteristic of main task subsequent, and also on the body segment which one wants to enhance, it might be necessary to perform different types of exercise such as CA in the same training session.

What would be a problem for sports that movements require multijoint and intersegment coordination to achieve higher levels of performance (eg, martial arts, football, rugby, Crossfit, and Olympic weightlifting [OWL]). For example, in Muay Thai, the fighter can strike punches, kicks, elbows, and knees, but if a PAPE protocol is performed with the bench press exercise as CA, there may be an increase in the performance of punches and elbows. On the other hand, the kicks and knees of this same fighter probably will not benefit in the same way. If the practitioners' target is to generate PAPE on both members, they need to perform 2 different CAs, one for UL, and another for LL. In this regard, Kilduff et al[20] used bench press as the CA to generate PAPE on bench press throw (BPT) and squat as the CA to generate PAPE on countermovement jump (CMJ) with professional rugby players. In fact, some studies have already tried to evaluate the effect of remote PAPE, executing a protocol with CA from UL in an attempt to leverage LL, however, the results were inconclusive. In the Cuenca-Fernández et al,[21] 4 CA protocols (only UL, only LL and UL + LL, and control) were compared to verify whether there would be an effect of PAPE on the performance of squat jump and power pushup simultaneously. Despite the observed positive effect on the general performance of these exercises ($P < .05$), thus confirming the remote effect of

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PAPE, there was a decrease in the impulse for the conditions only UL and UL + LL. This is one of the reasons why the authors point out the need for further studies to confirm the remote effect theory. Laskin et al[5] study investigated a bench press protocol as CA to generate PAPE in CMJ; however, there were no significant effects ($P > .05$). Perhaps, a simultaneous effect would be observed if a high-intensity exercise involving both UL and LL was applied. This could reduce the total time expended with the CA.

The possibility of using exercises derived from OWL to increase intensely muscle strength and power performance through PAPE in ULs and LLs simultaneously is attractive, improving, for example, CMJ and BPT performance.

However, studies applying different OWL exercises (eg, hang clean, snatch pull, clean pull, or power clean) have emphasized only one phase of the movement, more characteristic of UL or LL.[10,12,13,17,22,23] In the hang clean, for example, only the “pulling” phase of the clean and jerk (C&J) was executed (ie, knees and hips extension) and not the “pressing” phase (ie, elbow extension and shoulder flexion). Thus, LL is more requested to perform this type of movement. In addition to these considerations, no studies have evaluated the C&J as a form of CA for PAPE, nor whether this exercise (which contains the pulling and pressing phase) could increase performance in both UL and LL.

The empirical justification of coaches and researchers for not using the C&J as a CA to generate PAPE is that it is a very complex exercise to perform and manipulate heavy loads for practitioners of different sports other than strength. Thus, perhaps guaranteeing a minimum previous experience with C&J (eg, 3 y of training), associated with a certain level of strength (ie, perform 1RM with 1.15 times body mass, as proposed in past studies for power clean[10]), this type of CA could be used to generate PAPE regardless of the practitioner’s modality. In parallel, it would also be interesting to test several intervals between CA and the main activity for this new protocol model. Since the optimal potentiation range may depend on the type of CA, strength level of the practitioner, training experience, and their muscle fiber structure. Thus, there would be no need to run one CA for each limb with the added benefit of developing PAPE simultaneously in UL and LL. However, this assumption needs further studies to be confirmed.

Since a protocol capable of generating a PAPE effect in a nonlocalized/simultaneous way would have great applicability in the training of several sports, so far, no study has sought to evaluate a single exercise that integrates high performance in UL and LL in the same repetition as in the case of C&J. The aim of the present study was to determine whether the C&J could induce PAPE in both UL and LL performances. We also tested whether different intervals between C&J and the main task (4, 7, and 12 min) would influence the levels of PAPE. We hypothesized that the C&J would induce PAPE in both UL and LL performance, and PAPE would be greater using 7- and 12-minute intervals than with 4-minute intervals.

Methods

Subjects

Participants were selected through a nonprobabilistic sampling process and for convenience due to the high level of performance required to participate in the study. The inclusion criteria were: (1) male; (2) age between 18 and 45 years; (3) C&J 1RM equivalent to at least 1.15 times body mass at the time of sample selection and had a minimum of 3 years of experience with OWL; (4) without musculoskeletal injury, disease, or limitations that would prevent the performance of physical tests or any proposed exercise; and (5) not using medications affecting responses to exercise. The Institutional Research Ethics Committee approved this study (number 4.472.447). All participants received explanations about the procedures and risks and signed an Informed Consent Form prior to participation.

Design

The present study included 6 days of data collection divided into 2 stages. The first stage had the following objectives: (1) anthropometric measurements; (2) 1RM tests’ familiarization for the exercises: C&J, squat, and bench press; and (3) familiarization with CMJ and BPT tests. The second stage was the experimental sessions (Figure 1), in which participants randomly performed the C&J or control condition (48–72 h between them). A general warm-up consisting of 5-minute treadmill running at 8 km/h plus 5 minutes of participant’s self-selected preparatory exercises (such as stretching, mobility exercises, and other generic movements preceded each experimental condition). After the warm-up, participants executed 4 sets of C&J with 2 minutes of rest between them (first set 30%1RM, second set 50%1RM, third set 65%1RM, and fourth set 80%1RM), followed by CMJ and BPT tests performed 4, 7, and 12 minutes after the C&J. In these tests, 3 maximum attempts were performed with a 15-second interval between them. The control condition consisted of a general warm-up followed by CMJ and BPT tests performed 4, 7, and 12 minutes. The conditions and test sequence (CMJ and BPT) were randomly determined for each participant.

Anthropometric Measurements

Body mass (Filizola mechanical scale, Garmin Scale) and height (Sanny portable stadiometer, Sanny) were measured in the first

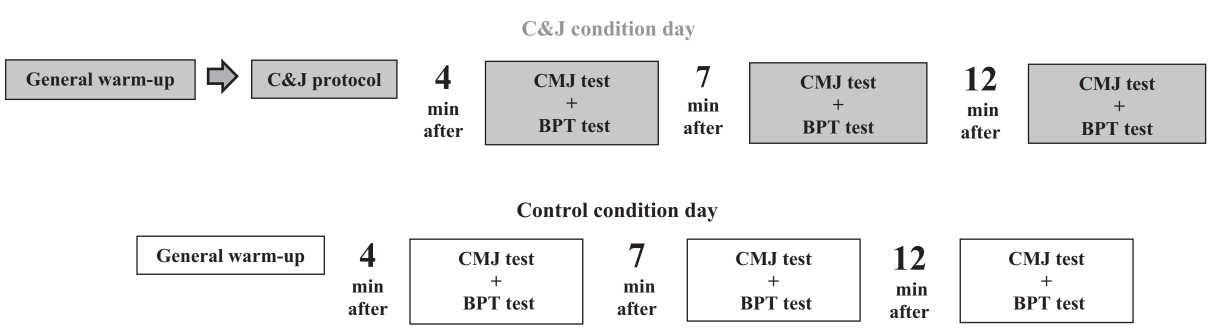

Figure 1 — Sequence of experimental conditions. *Note*. Condition and test order were random and balanced. BPT indicates bench-press throw; C&J, clean and jerk; CMJ, countermovement jump.

session. In both cases, the participants were barefoot and wearing only shorts. The body mass index value was obtained by body mass (in kilograms)/height^2 (in meters squared).

Maximum Dynamic Strength Tests for C&J, Squat, and Bench-Press Exercises

A standardized general warm-up (5-min treadmill running at 8 km/h) plus 5 minutes for the participant's usual preparatory exercises (stretching, mobility exercises, and other generic warm-up movements) preceded all 1RM testing sessions. To ensure reproducibility of participants' positioning on different testing sessions, adhesive tape was used to mark handle position on the bar, feet position on the floor, and torso position on the bench for the bench press. For the squat, an adjustable bench was positioned behind the participant to guarantee the same range of motion. All tests measured the maximum amount of weight that could be lifted in a complete movement cycle, and the 1RM value was determined with a maximum of 5 attempts with 3 minutes of rest between them. As a familiarization criterion, if the load variation between the last 2 testing days of a given exercise was >5%, a new session was scheduled.[24] For the C&J exercise, the 1RM test protocol followed the criteria proposed by Haff et al.[25] Participants started with a specific warm-up executing 3% × 30%, 3% × 50%, 1% × 70%, and 1% × 90% estimated RM with 2-minute rest between sets. The 1RM attempts started 4 minutes after the end of the specific warm-up. To validate a repetition in C&J, the International Weightlifting Federation's criteria were adopted.[26] Briefly, the maximal strength test consisted of raising the greatest amount of weight over the head with extended elbows in a 2-phase movement. The first phase was started with the weight bar on the floor. The participants would approach the bar and with their knees bent and hold the bar with both hands with a pronated grip, trying to keep their back upright and at 45^o in relation to the floor. From that position, they would have to raise the bar with a single movement up to their shoulders, resting it in that region for a few seconds before the second phase. In the second phase, the bar was raised above their head with elbow extension movement combined with a rapid hip and knee flexion/extension movement.

For the squat 1RM test, the protocol proposed by Paulo et al[27] was used. After the general warm-up, participants executed a specific warm-up with 5% × 50% and 3% × 70% estimated 1RM separated by 5-minute rest. The 1RM attempts started 3 minutes after the end of the specific warm-up. The participants performed a complete repetition, starting from the standing position with extended knees, bending their knees until their thighs were parallel to the floor, and the repetition finished returning to full extension. This sequence was performed with the bar positioned on the participant's back shoulders.

For the bench press 1RM test, the protocol proposed by Paulo et al[27] was also used. After the general warm-up, participants executed a specific warm-up with 5% × 50% and 3% × 70% estimated RM separated by 5-minute rest. The 1RM attempts started 3 minutes after the end of the specific warm-up. Participants lay on the bench in a supine position with shoulders under the bar and both feet on the ground. Participants then gripped the bar in a comfortable position. The distance of both thumbs from the center of the bar was recorded with a measuring tape placed over the bar. The repetition started in full elbow extension, the bar was lowered up to touching the chest, and the repetition finished, returning to full extension.

CMJ Test

The CMJ was performed following previous recommendations.[15] Briefly, each participant started from a standing position, with knees fully extended and hands on the hips. The participant then bent his knees and hips and rapidly extended them, jumping as high as possible in 3 trials separated with a 15-second interval between attempts. The participants were free to determine the amplitude of the downward part of the movement that would generate a higher performance. The CMJ was performed on the force platform (EMG System Biomec400-412). The CMJ with the highest height was selected for further analysis.

BPT Test

The BPT test was performed on the Smith machine, with a displacement sensor (EMG System SAS100V8-WF), coupled to the bar. This instrument records the displacement of the bar over time (sampling rate = 200 Hz) during the entire execution of the movement. The movement began with the elbows fully extended. Then, an eccentric action was performed until the bar touched the chest, followed immediately by a concentric action that launched the bar as high as possible. The load was set at 30%1RM. To ensure participants' positioning reproducibility, adhesive tape markings were made to register torso position on the bench, the distance between hands, and the position of the feet. Participants performed 3 attempts, with a 15-second interval between attempts. The attained bar height was established by subtracting the greatest height recorded from the initial bar position on the chest. The BPT with the highest height was selected for further analysis.

Statistical Analyses

The Statistical Package for the Social Sciences software (SPSS, version 20) was used for data analysis. Descriptive statistics (mean and standard deviation) were presented for the sample characterization variables (age, height, body mass, body mass index, 1RM on C&J, squat, and bench press) and dependent variables (maximum height in the CMJ and the BPT). The Shapiro–Wilk test confirmed the normality of the data. A repeated measures analysis of variance (ANOVA) was used to verify whether there was a significant difference in performance of the CMJ and the BPT between conditions. When F was significant ($P < .05$), the Bonferroni post hoc test was used to identify the difference between the means. A repeated measures ANOVA was used to verify whether there was a significant difference in performance in relation to the means obtained from the CMJ and the BPT between the conditions (C&J vs control) and time (4 vs 7 vs 12 min). A paired t test was also used to compare the higher values the participants obtained in each condition (C&J vs control). The effect size (ES) was used to verify the magnitude of the effect between C&J and control conditions using the formula proposed by Cohen where: mean C&J condition − mean control condition/standard deviation grouped; being adopted as grouped standard deviation the equivalent of: $\sqrt{[(DP_{cleanandjerkcondition}{}^2 + DP_{controlcondition}{}^2)/2]}$.[28] Taking into account that volunteers were highly trained,[29] the ES was considered trivial when <0.25, small when ≥0.25 and <0.50, moderate ≥0.50 and <1.0, and large when ≥1.0.

Results

The characteristics of the participants (age, body mass, height, body mass index, C&J, squat, and bench press 1RM) are detailed in Table 1.

There was a main effect of condition on CMJ height ($F = 5.550$; $P = .038$; power = 0.575). The CMJ height was higher in the C&J

condition than in the control condition (+3.66%, ES = 0.30 [small effect]; Table 2). There were no statistical differences between 4, 7, and 12 minutes (F = 2.088; P = .175; power = 0.331). Descriptively, the higher CMJ height performance among participants was found at 4 minutes (n = 7), followed by 12 minutes (n = 4) and 7 minutes (n = 1; Figure 2). Even when evaluating only the higher values obtained for each participant regardless of time, between conditions, the t test no showed a significant increase in the C&J condition compared with the control (+4.13%, t[11] = −2.034; P = .067).

There was main effect of condition on BPT (F = 5.471; P = .039; power = 0.569). The BPT height was higher in C&J condition than in the control condition (+4.44%, ES = 0.26 [small effect]; Table 3). There were no statistical differences between 4, 7, and 12 minutes (F = 1.462; P = .277; power = 0.242). Descriptively, the higher BPT height performance among participants was found at 7 minutes (n = 5), followed by 4 minutes (n = 4) and 12 minutes (n = 3; Figure 2). Even when evaluating only the higher values obtained for each participant regardless of time, between conditions, the t test no showed a significant increase in the C&J condition compared to the control (+4.12%, t[11] = −2.099; P = .060).

Discussion

The present study aimed to verify whether the C&J exercise would be able to produce a PAPE effect in both UL and LL. In parallel, it aimed to compare the effect of different rest intervals. The main findings were that, regardless of interval duration, the C&J condition resulted in a greater average height of CMJ and BPT than in the control condition.

Table 1 Sample Characterstics

	Mean (SD)
Age, y	30.17 (5.96)
Training experience, y	5.17 (1.66)
Height, m	1.75 (0.12)
Body mass, kg	83.18 (3.24)
BMI, kg/m^2	27.17 (3.24)
1RM C&J—absolute, kg	109.08 (14.87)
1RM C&J—relative body mass	1.32 (0.16)
1RM squat—absolute, kg	155.08 (26.50)
1RM squat—relative body mass	1.83 (0.34)
1RM bench press—absolute, kg	99.33 (20.62)
1RM bench press—relative body mass	1.24 (0.19)

Abbreviations: BMI, body mass index; C&J, Clean and Jerk; 1RM, 1-repetition maximum.

Our first hypothesis was confirmed because the repeated measures ANOVA shows that C&J execution resulted in +3.66% and 4.44% increase in CMJ and BPT performance, respectively, when compared with a control condition (P < .05). Since our aim was to compare different conditions along different intervals, the most appropriate statistical test would be the repeated measures ANOVA. However, apparently the effects of PAPE are dependent on individual factors and on different parts of the body.[2,3] For CMJ, the highest height performance was found in 58.3% of participants at 4 minutes, followed by 33.3% at 12 minutes and 8.3% at 7 minutes. As for the BPT, the highest height performance occurred in 41.7% of the participants in 7 minutes, followed by 33.3% in 4 minutes and 25% in 12 minutes. In this sense, selecting only the highest jump or highest throw in each condition, respectively, and the t test revealed a marginal P (P = .06). These findings agree with other studies involving CMJ that also observed PAPE effects after OWL CA.[12,17] For example, in the study by Mccann and Flanagan,[17] where improvements of +2.16% (P < .05; ES = 0.09) in the CMJ after a protocol involving hang cleans. Similarly, Chiu and Salem[12] demonstrated improvements of +5.90% (P < .05; ES = 1.75) applying snatch pull. However, the results do not agree with what was observed by the Dinsdale and Bissas study[13] where no significant effects of PAPE were found after the execution of the hang clean exercise. This may have happened because the sample in question had a lower level of strength (1RM hang clean = 1.02 × body mass) than recommended by Seitz and Haff.[10] In the present study, we attended this criterion selected by using only advanced practitioners with 1RM on C&J ≥ 1.15 × body mass.

To our knowledge, this is the first study investigating the effect of C&J on UL, so it is not possible to make a direct comparison with other studies. However, our findings show a PAPE similar or superior to other studies that used the bench press exercise to potentiate the BPT.[30,31] A possible explanation is that C&J might produce PAPE in the synergistic muscles that both exercises (C&J and BPT) share in common, such as pectoralis, triceps brachii, and deltoids.[32,33]

Contrary to our second hypothesis, we found no influence of interval duration on PAPE-induced effects of C&J on either CMJ or BPT. This finding contrasts with previous reviews,[2,3] suggesting that intervals between 5 and 7 minutes are optimal for PAPE purposes. The same happened in the study of Seitz and Haff,[10] where 7 minutes seems to have been the proper interval for PAPE with the power clean exercise. The optimal potentiation interval depends not only on the participant's strength level, but also on the particular exercise selected for CA (traditional, ballistic, isometric, and concentric), and the exercise, or sports skill that is being potentialized[6]—for example, BPT, 20-m sprint, and bench press. As the mentioned studies did not include the C&J as CA in their analysis, our findings indicate that in general, the interval between 4 and 12 minutes for the manifestation of PAPE is not essential

Table 2 Maximum Height on the Countermovement Jump (in Centimeters)

	Control condition (n = 12)	C&J condition (n = 12)	DIF% C&J/control	ES (95% CI)
4 min	33.21 (3.86)	35.27 (3.83)	+6.19%	0.53 (−1.40 to 0.38)
7 min	33.59 (4.45)	34.60 (3.97)	+3.00%	0.24 (−1.03 to 0.57)
12 min	33.30 (4.32)	33.90 (4.18)	+1.88%	0.14 (−0.94 to 0.66)
Combined values	33.37 (4.10)	34.59 (3.92)*	+3.66%	0.30 (−1.10 to 0.51)

Abbreviations: C&J, clean and jerk; DIF%, percentage difference between C&J condition compared with control; ES, effect size; 4 min, 4-minute interval; 7 min, 7-minute interval; 12 min, 12-minute interval.
*Significant difference in relation to control condition (P < .05).

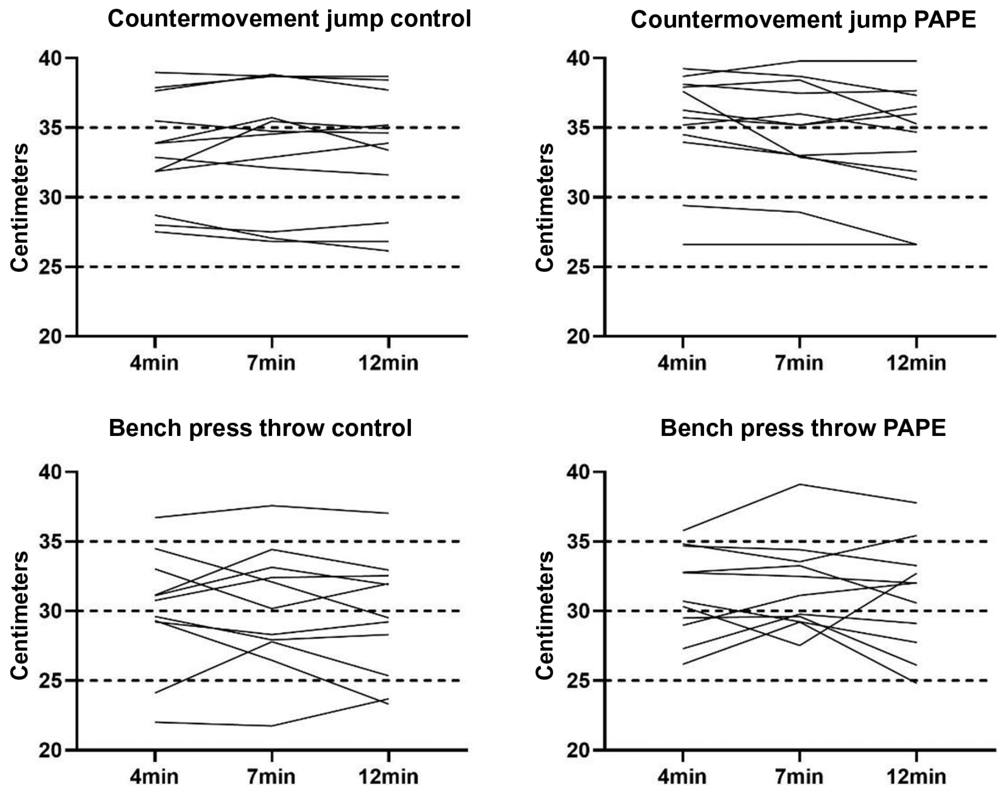

Figure 2 — Individual perfomance for bench-press throw and countermovement jump in both conditions. PAPE indicates postactivation performance enhancement.

Table 3 Maximum Throwing Height on the Bench-Press Throw (in Centimeters)

	Control condition (n = 12)	C&J condition (n = 12)	DIF% C&J/control	ES (95% CI)
4 min	29.16 (5.25)	30.21 (4.71)	+3.60%	0.21 (−1.01 to 0.60)
7 min	29.24 (5.30)	30.74 (4.69)	+5.15%	0.30 (−1.10 to 0.51)
12 min	28.73 (5.07)	30.04 (5.09)	+4.55%	0.26 (−1.05 to 0.55)
Combined values[a]	29.04 (5.06)	30.33 (4.70)*	+4.44%	0.26 (−1.06 to 0.55)

Abbreviations: C&J, clean and jerk; DIF%, percentage difference between C&J condition compared with control; ES, effect size; 4 min, 4-minute interval; 7 min, 7-minute interval; 12 min, 12-minute interval.
[a]Average value of the main effect of analysis of variance, regardless of time.
*Significant difference in relation to control condition ($P < .05$).

when the selected exercise is C&J and the main activities are CMJ or BPT ($P > .05$). However, when analyzing the ES between the CMJ intervals, apparently the 4-minute time seems to have produced greater effects on jump height (ES = 0.53) compared with ES = 0.24 and ES = 0.14 for 7 and 12 minutes, respectively. Thus, we understand that although there is simultaneous potentiation, the most suitable potentiation interval for optimal interlimb response may be different.

Some limitations of the present study should be acknowledged. Our study used a high-level sample in the OWL. Future studies should assess the effectiveness of PAPE in practitioners of different levels. Another limitation refers to the power of the sample since it was below 60%. This study was conducted in the middle of the COVID-19 pandemic, thus making access difficult for practitioners who had strength levels compatible with what was proposed in the inclusion criteria. On the other hand, this number of participants (n = 12) is in line with that used in other PAPE studies (9–18 participants).[13,15,18,30,31]

We did not perform jumps and throws before the experimental session on the day because there is evidence that these previous activities could already have generated a PAPE.[3,16,34] This would be one more intervening variable in addition to the general warm-up to identify the exclusive effect from C&J. Because the analysis would be the sum of the general warm-up + jump pre or throw pre + C&J protocol in the experimental condition versus general warm-up + prejump or prethrow in the control condition. Therefore, this experimental presession value was discarded. Also, baseline values were not performed 1 day before or 3 hours before the session. Future studies could adopt this measure in their design. Also, in our study, the potentiation intervals were tested on the same day, for each of the conditions. Future studies could perform the tests of different intervals, in alternating sessions.

Practical Applications

The C&J exercise can be used as CA for technical movements that require the synergistic integration of both UL and LL (eg, power clean, power snatch, drop snatch, split jerk, and hang snatch pull) or, can also optimize traditional and ballistic strength training exercises commonly performed by weightlifters as supplementary training work (eg, depth jumps, jump squats, sprints, deadlifts, front squats to develop strength and power of LL, and plyometric push-ups, medicine ball throw, bench press, shoulder press, upright rows, lateral raises, and dips as UL exercises' development). It can also be used by Crossfit and Strongman athletes who perform strength exercises very similar to weightlifting in their training and have strength levels greater than, or equal to, the participants in this study. It is also worth remembering that the use of weightlifting exercises as a form of strength training can also bring chronic benefits in terms of sports performance in several other sports, for example, running, handball, swimming, rowing, football, and Muay Thai. Therefore, carrying out a training process (involving improvement of technical and performance attributes in these exercises) could bring double benefits, in the form of an acute and chronic effect, depending on the protocol and the form of use during training.

Conclusions

In conclusion, this study may have important implications in weightlifting training and competitions, as the clean-and-jerk exercise can generate postactivation performance enhancement simultaneously in both the upper and lower limbs, regardless of intervals between the conditioning activity and the main task.

Acknowledgments

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